

Cloud Computing & DevOps – AWS Handbook

SESSION 5

Database Fundamentals & Amazon RDS

Building Cloud-Based Database Solutions

Duration	2 Hours
Prerequisites	AWS Account, Basic SQL Knowledge
Learning Outcome	Understand database fundamentals, create an Amazon RDS instance, connect to a database and execute basic SQL operations.

Why Are We Learning This?

Most applications require a database to store and retrieve information. Amazon RDS simplifies database administration by providing a managed database service in AWS.

Architecture Diagram



Key Concepts

Database	Organized collection of data
RDBMS	Relational Database Management System
SQL	Language used to query databases
Amazon RDS	Managed database service
Table	Stores records in rows and columns

Practical Activities

- Create an Amazon RDS instance
- Configure database connectivity
- Connect to the database
- Create tables and records
- Execute SQL queries

Expected Outputs

RDS instance running, successful connection, tables created, records inserted and queries executed.

What You Should See

Paste Screenshot Here

Amazon RDS Dashboard

Paste Screenshot Here

Database Instance Details

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Database Connection Screen

Paste Screenshot Here

Table Creation Output

Paste Screenshot Here

SQL Query Results

Journal Writing Template

Aim

To create and access a relational database using Amazon RDS.

Theory

Amazon RDS provides managed relational databases.

Procedure

Write the steps performed during the practical.

Observations

Database created and SQL operations completed successfully.

Conclusion

Amazon RDS was successfully used to manage a cloud database.

Viva Preparation

1. What is a Database?
2. What is SQL?
3. What is Amazon RDS?
4. What is an RDBMS?
5. Benefits of managed databases?

Troubleshooting Box

Problem	Possible Cause
Cannot connect	Security group issue
Login failed	Wrong credentials
Query error	SQL syntax issue
Instance unavailable	RDS not running

Industry Relevance

Amazon RDS is widely used for enterprise applications, e-commerce platforms and cloud-native workloads.

Session Summary

Students learned database fundamentals and practical usage of Amazon RDS.